HelpDesk AI (OmniDesk) — Complete System, SLA & Viva Guide

Comprehensive Plain-English Guide: Business Problem, System Operation, SLA Mathematics, Escalation Engine & Interview Q&A;

Author: Md. Asif Iqbal | **Purpose:** Personal Mastery, Architecture Defense & Viva Preparation | **Tech Stack:** Next.js 16 + Express + MongoDB Atlas + Groq/Gemini

1. Executive Plain-English Summary: What Problem Does This Project Solve?

Imagine a mid-to-large software company with 50,000 customers. Everyday, hundreds of support inquiries flood their customer inbox: *'Payment failed!'*, *'How do I reset my password?'*, *'Critical database deadlock in production!'*. In traditional companies, human support agents manually read each message one by one, guess who should handle it, and spend hours drafting basic replies. Consequently, frustrated customers wait 12+ hours, cancel subscriptions, and cause severe revenue churn.

HelpDesk AI (OmniDesk) transforms this chaotic, error-prone process into an automated, AI-governed support desk:

- **Real-Time NLP Triage (Groq + Gemini):** The exact moment a customer finishes typing, multimodal LLMs evaluate the message in milliseconds. The system scores urgency (0–100), measures customer emotional sentiment (-1.0 to +1.0), and classifies the issue into technical, billing, or account domains.
- **Algorithmic SLA Timers:** Every ticket is locked to an immutable resolution deadline (Urgent: 2h, High: 8h, Medium: 24h, Low: 48h) with real-time dynamic countdowns.
- **Autonomous Churn Escalation:** If customer distress exceeds safe thresholds (Sentiment ≤ -0.6) or if an urgent ticket remains unhandled, the system autonomously escalates the ticket to Tier-2 senior staff.
- **AI Copilot Suggested Responses:** When a support agent opens the ticket, an AI-drafted reply is already prepared. The agent clicks 1 button to insert and dispatch it, slashing First Response Time (FRT) by over 75%.

2. Complete Deep-Dive: What is an SLA (Service Level Agreement)?

SLA stands for **Service Level Agreement**. In the software and enterprise SaaS industries, an SLA is a formal, legally binding commitment between a service provider and its clients defining the maximum acceptable time to acknowledge and completely resolve an issue. If a company breaches an SLA, it frequently incurs contract penalty refunds, loss of enterprise clients, and permanent brand damage.

The Two Primary SLA Time Metrics:

- **First Response Time (FRT):** The duration from ticket creation until an agent or AI dispatches the first substantive, personalized reply to the customer.
- **Mean Time to Resolution (MTTR):** The total time elapsed from ticket submission until the ticket status transitions to **RESOLVED**.

System SLA Tier Matrix & Enforcement Rules:

Urgent	2 Hours	< 15 mins	Production down, payment double-charge, security breach	Auto-escalate to Tier-2, immediate Red badge, priority queue top
High	8 Hours	< 1 hour	Major feature broken for multiple users, reporting failure	Amber warning badge at 4h mark, alert assigned agent
Medium	24 Hours	< 4 hours	Non-blocking bugs, UI layout misalignment, single-user lag	Standard triage queue, periodic status recalculation
Low	48 Hours	< 8 hours	General inquiries, documentation clarity, feature feedback	Batch queue review, auto-archive after resolution

How SLA is Calculated in Code:

slaDeadline = new Date(ticket.createdAt.getTime() + priorityHours * 3600 * 1000)

remainingTime = slaDeadline - Date.now()

- **Healthy (Green):** > 50% SLA duration remaining.
- **At Risk / Warning (Amber):** < 50% SLA duration remaining.
- **Breached (Red):** remainingTime ≤ 0 while status is not RESOLVED or CLOSED.

3. What is Ticket Escalation (■■■■■■■■)? Architecture & Triggers

Ticket Escalation is the operational mechanism of transferring an unhandled, high-risk, or technically complex ticket from Tier-1 frontline support agents to Tier-2 senior engineers, database administrators, or department managers. Escalation ensures that high-impact issues receive immediate executive focus before they cause customer churn or legal breach.

The Two Escalation Paths in OmniDesk AI:

1. Autonomous AI & Algorithmic Escalation:
• *Sentiment Distress Trigger:* If the NLP triage engine detects high customer distress (Sentiment ≤ -0.6 or Urgency ≥ 85) upon submission, the ticket is immediately escalated to Tier-2.
• *SLA Threat Cron:* A scheduled background engine continuously audits open tickets. If an urgent ticket consumes more than 50% of its SLA without agent pickup, it triggers automated escalation.

2. Manual Staff Escalation:
Any support agent or administrator can click **'Escalate Ticket'** in the ticket detail view. The staff member submits a structured reason (e.g., *'Database replication failure requires DevOps patch'*). The ticket status is set to ESCALATED and priority is upgraded to Urgent.

The Immutable Escalation Audit Log (EscalationLog):

Every escalation action is permanently stored in MongoDB Atlas within the EscalationLog collection:
{ ticketId: ObjectId, escalatedBy: ObjectId, reason: String, previousPriority: String, newPriority: String, createdAt: Date }
This guarantees complete legal transparency, accountability, and post-incident compliance audits.

4. Public Response vs. Private Internal Note: Why Both Are Essential

In enterprise customer support, internal collaboration must be strictly separated from public client communication:

Feature Attribute	Public Response (Customer-Facing)	Internal Note (Private Team Staff)
Audience	The Customer and all internal support staff.	Support Agents and Administrators ONLY.
Customer Visibility	**100% Visible** on customer dashboard timeline.	**Completely Hidden** from customer portal & API responses.
Primary Purpose	Providing solutions, asking for debug info, status updates.	Sharing server logs, customer credit risk, sensitive bug reports.
Visual Badge in UI	Green / Slate bubble labeled 'Public Reply'	Yellow amber card labeled 'Internal Note (Staff Only)'

5. AI Multi-Model Orchestration: Why Groq + Gemini + Regex Fallback?

Groq (GPT-OSS-20B)	AI Copilot Suggested Reply generation.	Ultra-fast (> 500 tokens/sec)	Sub-second reply generation prevents agent hesitation.
Google Gemini 3.8 Flash	Real-time sentiment scoring & category classification.	Fast multimodal (1–2 sec)	Deep contextual reasoning for emotional distress detection.
Heuristic Regex Fallback	Keyword-driven triage (crash, payment, down).	Instant (< 1 ms offline)	Zero external dependencies guarantees 100% uptime.

6. Granular Role-Based Access Control (RBAC): 3 Isolated Perspectives

User Role	Allowed Operational Capabilities	Restricted Actions (Forbidden)
Customer	Submit tickets with live AI analysis; view personal tickets; monitor live SLA countdown timers; read and post public replies.	Cannot view other users' tickets; cannot view internal notes; cannot re-assign tickets or modify SLA deadlines.
Support Agent	Claim unassigned tickets ('Assign to Me'); dispatch 1-click AI suggested replies; compose private internal notes; mark tickets as Resolved.	Cannot delete tickets or edit user credentials; cannot bypass system SLA logs.
Administrator	View executive SLA compliance statistics (% breached, average resolution time); re-assign tickets between agents; audit complete escalation timeline.	Full system oversight with audited administrative logging.

7. System Architecture: How Frontend, Backend & Database Connect

- **Next.js 16 Frontend (Port 3000):** Built with App Router and Turbopack. Integrates Next.js rewrites (/api/:path* -> http://localhost:5001/api/:path*) to eliminate browser CORS preflight overhead and ensure zero-latency routing.
- **Express.js Backend API (Port 5001):** Decoupled Node.js REST server using TypeScript. Enforces Zod schema validation on every request, with centralized async error handling middleware.
- **MongoDB Atlas Persistence:** Cloud Replica Set M0 cluster. Indexed on slaDeadline, status, and priority for instant query sorting across thousands of concurrent tickets.
- **Dual Authentication:** Firebase Auth handles Google Sign-In Popup and Email/Password registration. The backend verifies credentials and issues a custom HMAC-SHA512 signed JWT authorization token for complete stateless security.

8. Viva & Technical Interview Cheat Sheet (Top 8 Winning Answers)

Q1: What is an SLA, and how does your application prevent SLA breaches?

Answer: SLA (Service Level Agreement) is the guaranteed contractual deadline to resolve an issue. Our system calculates an immutable deadline at ticket creation based on priority (Urgent: 2h, High: 8h, Medium: 24h, Low: 48h). To prevent breaches, dynamic color-coded countdown timers alert agents when less than 50% time remains, and our autonomous escalation engine automatically elevates urgent unassigned tickets to Tier-2 before the deadline expires.

Q2: How does the AI Copilot help the support agent in practice?

Answer: When an agent opens an inquiry, Groq (GPT-OSS-20B) has already generated a high-quality, empathetic response draft in sub-second time. With a single click ('Use Draft in Reply'), the draft is inserted into the editor. The agent can tweak it and dispatch it immediately, reducing First Response Time (FRT) from hours to seconds.

Q3: Why did you implement both Public Replies and Internal Notes?

Answer: In professional support, staff must collaborate internally without alarming the customer. Internal Notes allow agents to post server stack traces, database IDs, and customer churn risk scores visible only to colleagues. The customer's portal API strictly filters out internal notes, protecting internal company privacy.

Q4: What happens if Gemini or Groq APIs experience network outages?

Answer: We built a zero-dependency Heuristic Regex Engine as a resilient fallback. If cloud AI APIs fail, time out, or hit rate limits, the heuristic analyzer matches critical keywords ('crash', 'down', 'payment', 'fraud') and accurately classifies the ticket, ensuring 100% system availability.

Q5: How does your application authenticate users securely?

Answer: We implement a dual-layer authentication architecture. Firebase Authentication handles frontend Google Sign-In popups and email validation. Upon successful authentication, our backend synchronizes the user profile into MongoDB Atlas and generates a 512-bit signed Node.js JWT bearer token for API authorization.

Q6: Why are the frontend and backend deployed as two independent repositories?

Answer: Decoupling frontend and backend allows independent scalability, isolated CI/CD deployment pipelines, distinct dependency trees, and modular security perimeters. The frontend can be hosted on Vercel edge networks while the backend runs on containerized Node.js instances.

Q7: How is ticket escalation audited for compliance?

Answer: Every escalation event triggers an immutable write to the MongoDB EscalationLog collection, recording the ticket ID, actor ID, reason, previous priority, new priority, and exact timestamp. Administrators can review this un-editable audit history directly from the dashboard.

Q8: How did you ensure 100% responsiveness from 320px to 1440px+?

Answer: We used Tailwind CSS v4 fluid flex/grid wrappers, responsive padding, mobile drawer navigation, and stacked metric cards in AiInsightBox to eliminate any text clipping or horizontal scrolling across all mobile, tablet, and desktop viewports.